

```
from google.colab import files
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```



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Saving train.csv to train.csv

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load Titanic dataset
df = pd.read_csv("train.csv")

# Set plot style
sns.set(style="whitegrid")

# Show first 5 rows
df.head()
```



	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
# Data info
df.info()

# Summary statistics
df.describe()

# Check for missing values
df.isnull().sum()

# Value counts for categorical variables
df['Survived'].value_counts()
df['Sex'].value_counts()
df['Pclass'].value_counts()
```



<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
Column Non-Null Count Dtype
--- -
0 PassengerId 891 non-null int64
1 Survived 891 non-null int64
2 Pclass 891 non-null int64
3 Name 891 non-null object
4 Sex 891 non-null object
5 Age 714 non-null float64
6 SibSp 891 non-null int64
7 Parch 891 non-null int64
8 Ticket 891 non-null object
9 Fare 891 non-null float64
10 Cabin 204 non-null object
11 Embarked 889 non-null object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

count

Pclass	
3	491
1	216
2	184

dtype: int64

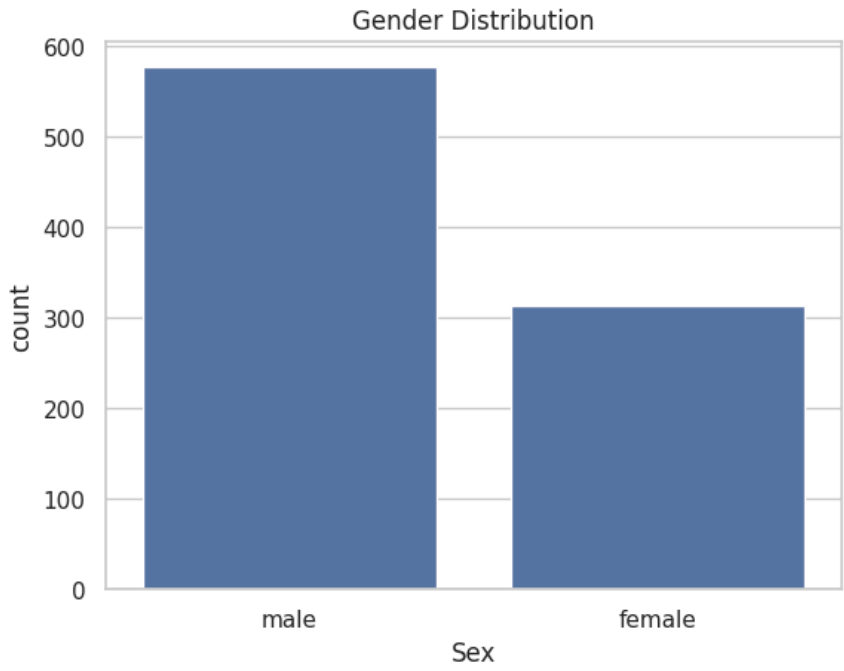
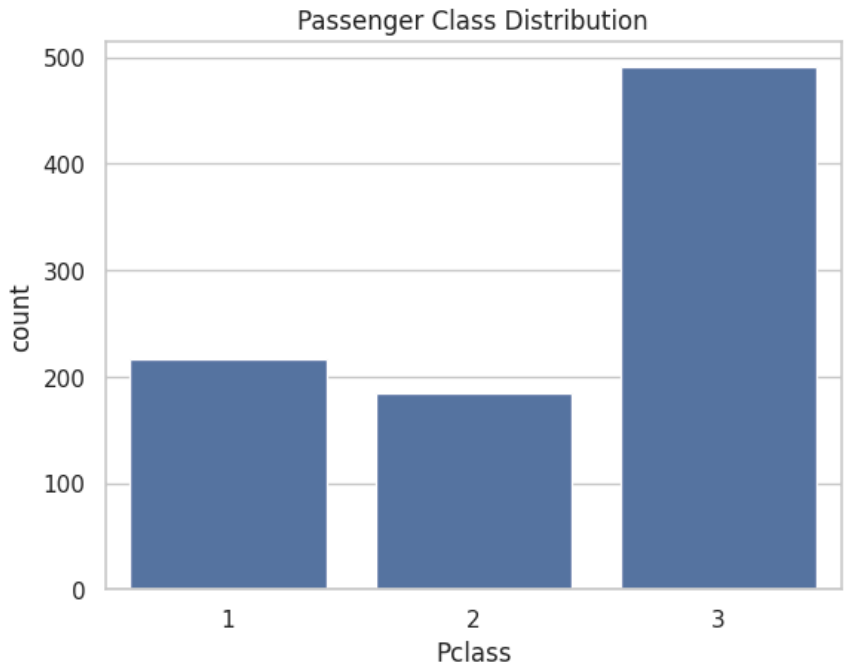
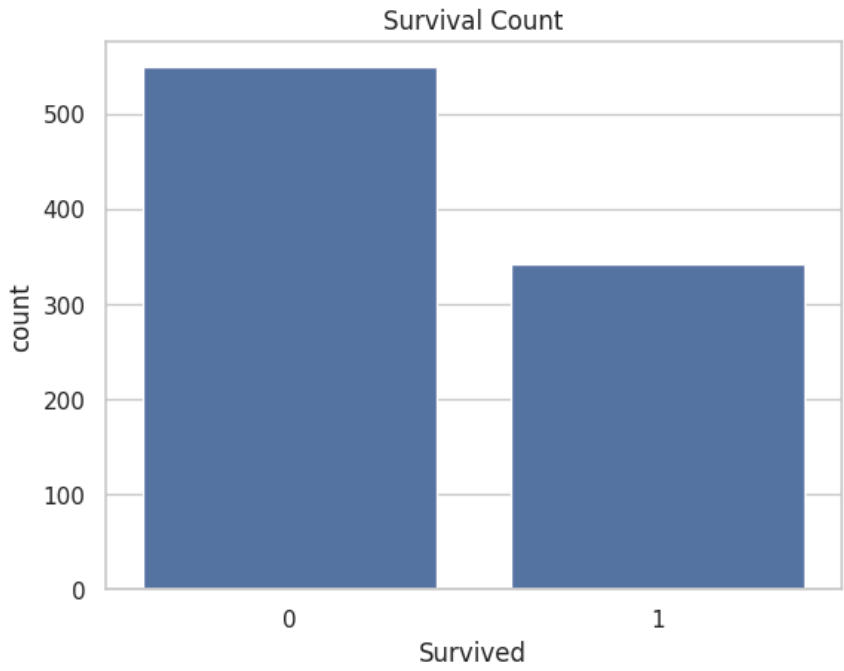
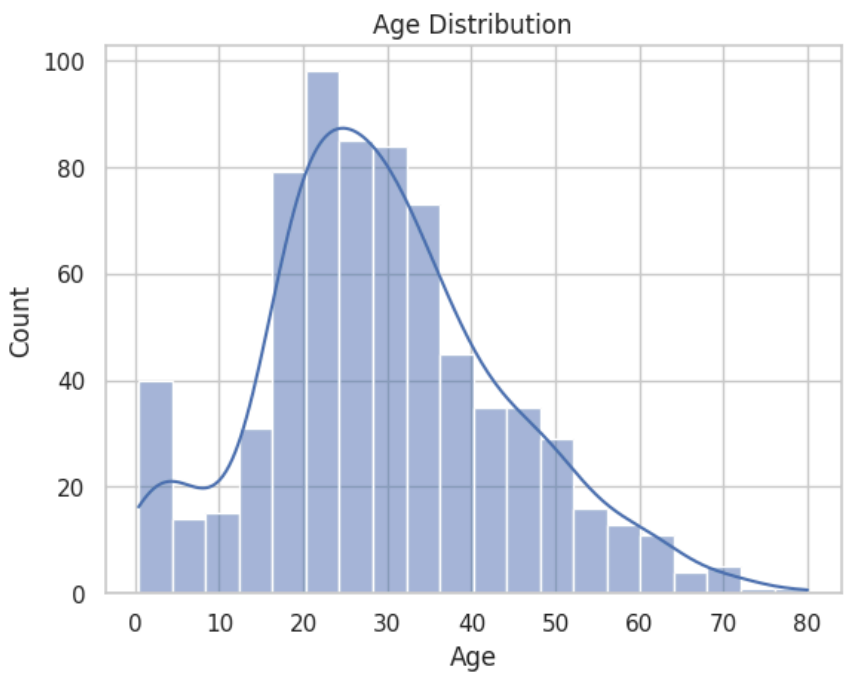
```
# Univariate Analysis

# Histogram of Age
sns.histplot(df['Age'].dropna(), kde=True)
plt.title('Age Distribution')
plt.show()

# Countplot for Survived
sns.countplot(x='Survived', data=df)
plt.title('Survival Count')
plt.show()

# Countplot for Passenger Class
sns.countplot(x='Pclass', data=df)
plt.title('Passenger Class Distribution')
plt.show()

# Gender distribution
sns.countplot(x='Sex', data=df)
plt.title('Gender Distribution')
plt.show()
```



Here, the observation of each graph ;

1. Age Distribution: The passenger age distribution is right-skewed with peaks in the younger to mid-adult range.
2. Survival Count: More passengers in the training data did not survive compared to those who did.
3. Passenger Class Distribution: The majority of passengers belonged to the 3rd class.

4. Gender Distribution: There were significantly more male passengers than female passengers

```
# Bivariate Analysis

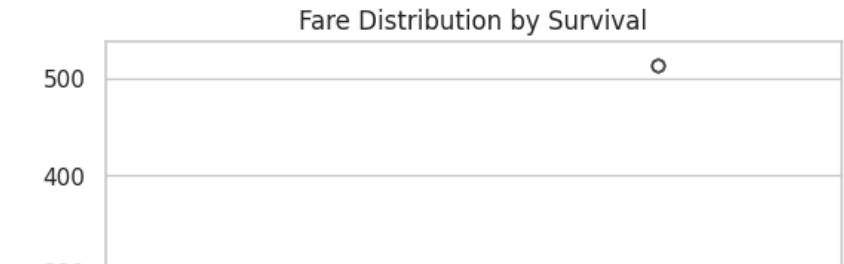
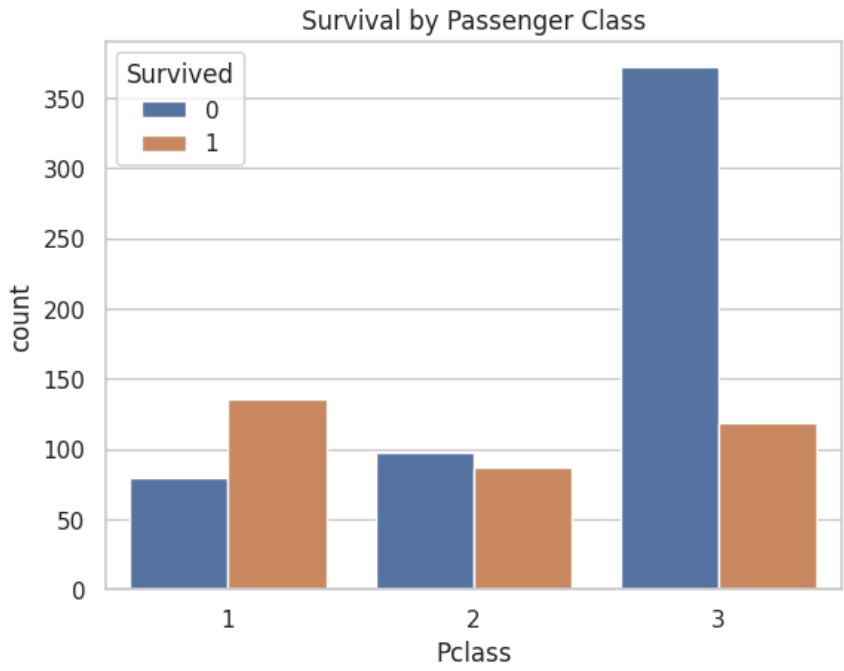
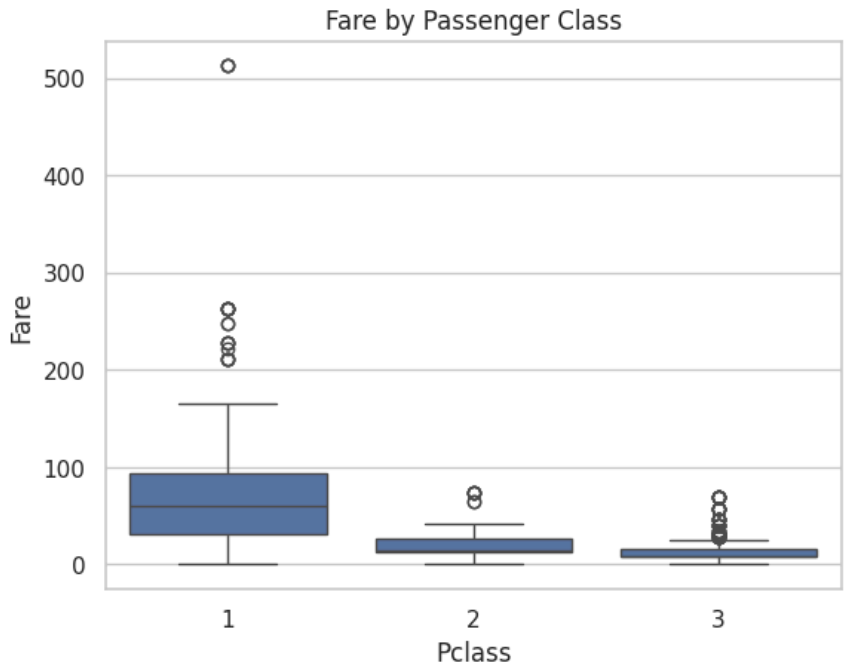
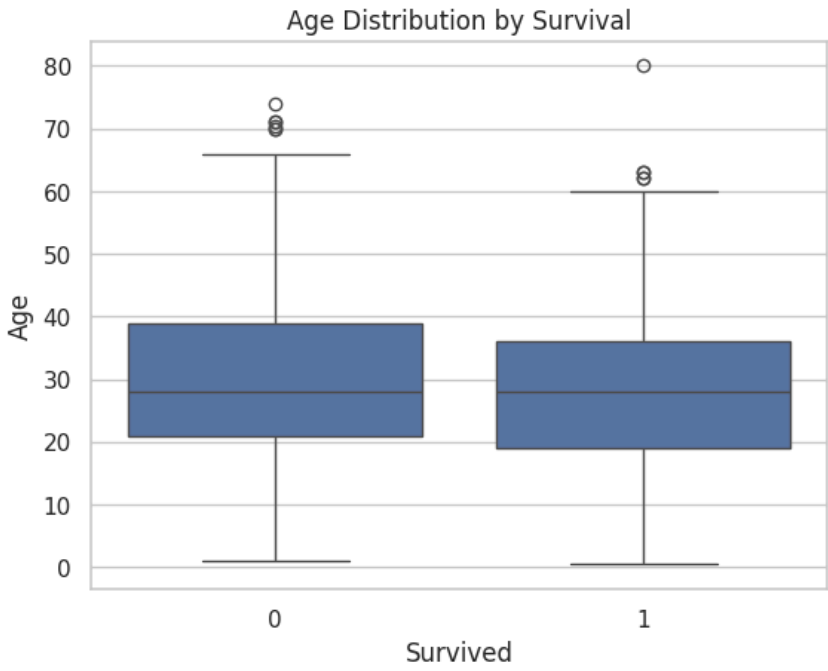
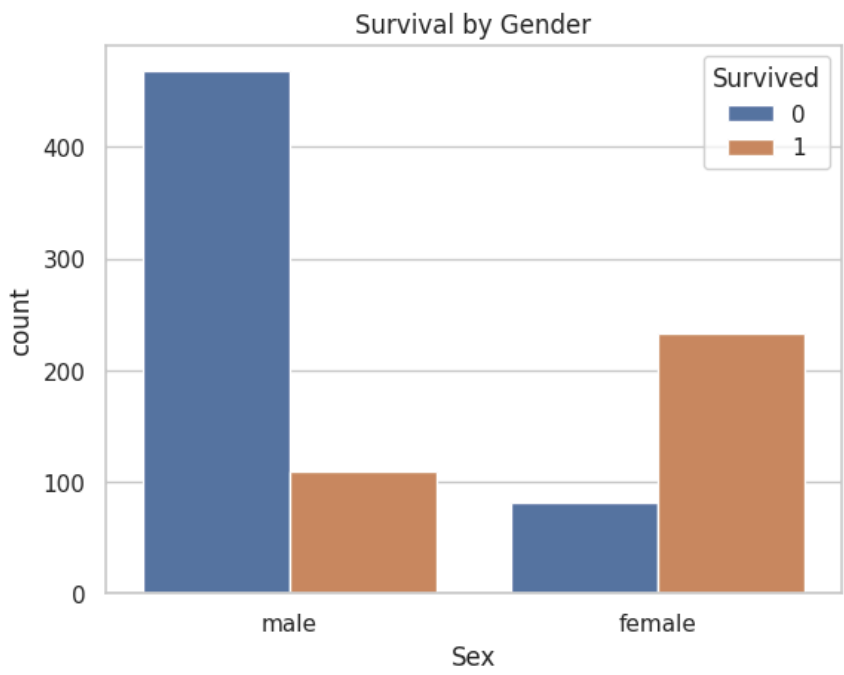
# Survival by Sex
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.show()

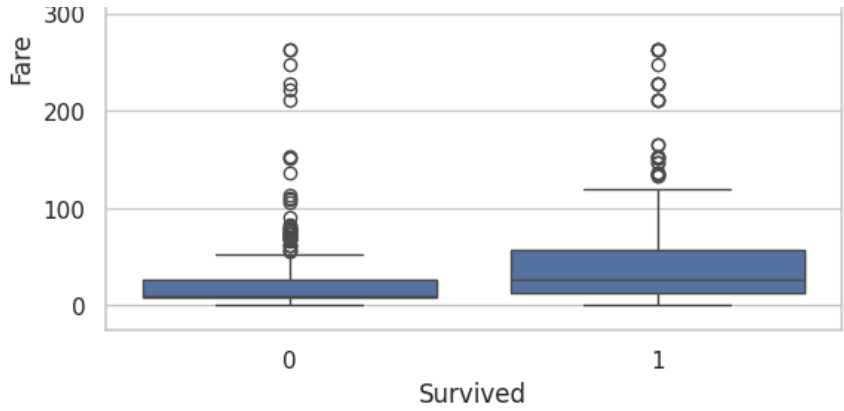
# Boxplot: Age vs Survived
sns.boxplot(x='Survived', y='Age', data=df)
plt.title('Age Distribution by Survival')
plt.show()

# Fare distribution by class
sns.boxplot(x='Pclass', y='Fare', data=df)
plt.title('Fare by Passenger Class')
plt.show()

# Survival by Pclass
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.show()

# Boxplot: Fare vs Survived
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival')
plt.show()
```





Here, the observation of each graph ;

- 1. Survival by Gender: Females had a substantially higher survival rate than males.
- 2. Age Distribution by Survival: Survivors and non-survivors had similar median ages, but the non-survival group had more older outliers.
- 3. Fare by Passenger Class: Ticket fare increased with higher passenger class.
- 4. Survival by Passenger Class: First-class passengers had the highest survival rate, while third-class had the lowest.
- 5. Fare Distribution by Survival: Survivors generally paid higher fares than non-survivors.

```
# Heatmap of Correlations
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()

# Pairplot (optional)
sns.pairplot(df[['Survived', 'Pclass', 'Age', 'Fare']], hue='Survived')
plt.show()
```

